

ASSIGNMENT 8

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3_98823_pranoti@localhost> select * from orders
-> where cnum =(select cnum from customers where cname ='cisneros');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

2. Extract all the orders of Motika.

```
KD3_98823_pranoti@localhost> select * from orders
-> where snum =(select snum from salespeople where sname ='MOTIKA');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

3. Find all the orders of salespeople servicing customers in London.

```
KD3_98823_pranoti@localhost> select * from orders
-> where cnum IN(select cnum from customers where city ='LONDON');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

4. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD3_98823_pranoti@localhost> select * from orders
-> WHERE amt >ANY(SELECT amt from orders o INNER JOIN customers c ON o.cnum =c.cnum
-> where city ='LONDON');
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

```
6 rows in set (0.00 sec)
```

5. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD3_98823_pranoti@localhost> SELECT * FROM customers
-> where rating>= ANY(select rating FROM customers c INNER JOIN salespeople s ON s.snum =c.snum
-> where sname ='Serres');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

```
4 rows in set (0.02 sec)
```

6. Select customers who have a greater rating than any other customer in Rome.

```
ERROR 1054 (42S22): Unknown column 'c.rating' in 'field list'
KD3_98823_pranoti@localhost> select * from customers
-> where rating>ANY(select rating from customers where city ='Rome');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

```
4 rows in set (0.00 sec)
```

7. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD3_98823_pranoti@localhost> select * from orders
-> where amt >ANY(select amt from orders where odate ='1990-10-04');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
8 rows in set (0.02 sec)
```

8. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3_98823_pranoti@localhost> select * from orders
-> where amt < ANY(select amt from orders o INNER JOIN customers c ON o.cnum=c.cnum
-> where city ='San jose');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

9. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD3_98823_pranoti@localhost> select o.amt ,c.cname ,c.rating from orders o
-> INNER JOIN customers c ON c.cnum =o.cnum
-> where amt >(select AVG(amt) from orders);
+-----+-----+-----+
| amt   | cname   | rating |
+-----+-----+-----+
| 5160.45 | Liu     | 200    |
| 9891.88 | Clemens | 100    |
| 4723.00 | Clemens | 100    |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

10. Select those customers whose rating are higher than every customer in London.

```
KD3_98823_pranoti@localhost> select * from customers
-> where rating >ALL (select rating from customers where city ='London');
+-----+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin  | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

11. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
ERROR 1054 (42S22): unknown column 's.snum' in 'where clause'
KD3_98823_pranoti@localhost> select snum,sname ,city from salespeople s
-> where city<>ALL(select city from customers c
-> where c.snum = s.snum);
+-----+-----+-----+
| snum | sname   | city   |
+-----+-----+-----+
| 1004 | Motika  | London |
| 1007 | Rifkin  | Barcelona |
| 1003 | Axelrod | New York |
+-----+-----+-----+
3 rows in set (0.02 sec)
```

12. Find names and numbers of all salesperson who have more than one customer.

```
ERROR 1054 (42S22): unknown column 'count' in 'having clause'
KD3_98823_pranoti@localhost> select s.snum,s.sname from salespeople s
-> INNER JOIN customers c ON s.snum =c.snum
-> GROUP BY s.snum,s.sname
-> HAVING COUNT(c.cnum)>1;
+-----+-----+
| snum | sname   |
+-----+-----+
| 1001 | Peel    |
| 1002 | Serres  |
+-----+-----+
2 rows in set (0.03 sec)
```

13. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3_98823_pranoti@localhost> SELECT snum, sname, city,
->      (SELECT COUNT(cnum)
->      FROM customers c
->      WHERE c.snum = s.snum) AS customer_count
-> FROM salespeople s
-> WHERE snum IN
-> (
->     SELECT snum
->     FROM customers
->     GROUP BY snum
->     HAVING COUNT(cnum) > 1
-> );
```

snum	sname	city	customer_count
1001	Peel	London	2
1002	Serres	San Jose	2

2 rows in set (0.02 sec)

14. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD3_98823_pranoti@localhost> SELECT snum, SUM(amt) AS total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > (
->     SELECT MAX(amt)
->     FROM orders
-> );
```

snum	total_amount
1001	15382.07

1 row in set (0.00 sec)